Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/08/2007
PI Name:	Caiozzo, Vincent J. Ph.D.		
Project Title:	Hypergravity Resistance Training: Countermeasure to Microgravity		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Muscle Alterations and Atrophy Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	vjcaiozz@uci.edu	**Fax:**	FY 949-824-3278
PI Organization Type:	UNIVERSITY	**Phone:**	949-824-5571
Organization Name:	University of California, Irvine		
PI Address 1:	Medical Sciences Bldg I, Room B152		
PI Address 2:			
PI Web Page:			
City:	Irvine	**State:**	CA
Zip Code:	92697-0001	**Congressional District:**	48
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	04/01/2004	**End Date:**	03/31/2008
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Baldwin, Kenneth (University of California, Irvine) Adams, Gregory (University of California, Irvine) Cavanagh, Peter (The Cleveland Clinic Foundation) Kreitenberg, Art (University of California, Irvine)		
Grant/Contract No.:	NCC 9-58-MA00403		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A program priority of NASA's Biomedical Research and Countermeasures Program (NRA-03-OBPR-04) is to determine the potential usefulness of artificial gravity as a countermeasure, especially with respect to skeletal muscle atrophy and loss of muscle function. The current proposal is a "proof-of-principle" of a unique countermeasure technology referred to as the "Space Cycle." The Space Cycle is a human powered centrifuge that can be used to generate various levels of artificial/hypergravity. Artificial gravity/hypergravity in a microgravity environment could be used as a novel method of performing resistance training under high loading conditions. The novelty of artificial gravity/hypergravity resistance training is that each element of the body is loaded proportional to the local gravitational field, and under hypergravity conditions muscles like those of the leg can be made to work against very high loads (e.g., + 2 body weights) without the need for external weights. The primary objective of this proposal is to use the Space Cycle to address the following general hypothesis: Artificial gravity can be used as a unique resistance training modality that acts as an effective countermeasure, preventing the loss of muscle mass and function that occurs due microgravity. In addressing this issue, a logical sequence of experiments is proposed with the following objectives: i) determine if squats performed under hypergravity conditions and without external weights can produce foot forces similar to those seen when performing squat resistance training (SRT) under normal 1 G conditions; ii) determine if squats performed under hypergravity conditions produce muscle adaptations similar to those seen using a squat resistance training program under normal 1 G conditions; and iii) determine if squat hypergravity resistance training (SHRT) program is an effective countermeasure to simulated microgravity. For the purposes of this grant proposal, we are focusing on SRT because squats recruit a broad spectrum of muscles in the leg and back, and are one of the classical exercises used by bodybuilders and athletes to hypertrophy muscles of the leg. Additionally, the so-called antigravity muscles of the leg are at the greatest risk for atrophy induced by microgravity. Furthermore, squats are a target exercise performed by astronauts on the International Space Station. During the second year of funding, the primary focus was on four goals: 1) continued evolution of the Space Cycle team; 2) continuing modifications and instrumentation of the Space Cycle; 3) integration of equipment for measuring foot forces, EMG, and joint angle; and 4) IRB modifications; 4) monthly team conferences; and 5) the performance of two studies. A considerable amount of time was spent addressing IRB issues. During the past year, two studies were performed. The first study represents the initial step in testing the hypothesis that hypergravity can be used as a unique modality for resistance training, maintaining the health and function of skeletal muscle in microgravity. The primary objectives of this study were: i) to determine if subjects could perform squats under hypergravity conditions without developing motion sickness or illusory motion; ii) to measure foot forces while performing squats under hypergravity conditions; and iii) to determine the power required by the cyclist to produce 1.5, 2.0, 2.5, and 3.0 Gz. All studies were performed using a unique human-powered short-arm centrifuge, the Space Cycle. One of the centrifuge arms of the Space Cycle had a gondola or cage like configuration that allowed subjects to perform squats. Subjects (male n = 20; and female n = 16) performed squats under hypergravity conditions (1.5, 2.0, 2.5, and 3.0 Gz), and during these tests foot forces were monitored using the Pedar-X system. Foot forces measured under the hypergravity conditions were normalized to foot forces measured at 1 Gz, and these were referred to as relative foot forces (RFF). All of the subjects were able to perform the hypergravity squats without developing motion sickness, and illusory motion was minimized by having the subjects fix their sight on the leading edge of the base plate. All of the male and female subjects were able to perform squats at 1.5, 2.0, and 2.5 Gz. However, two male and two female subjects were not able to perform squats at 3.0 Gz because the loading at this hypergravity condition exceeded their strength. The mean relative foot forces at the highest Gz (i.e., 3.0) were 2.3 and 2.5 times greater than body weight for the male and female subjects, respectively. The work rate required to power the Space Cycle was a linear function of Gz, and is well within the aerobic scope of most untrained individuals. In conclusion, the findings of this study demonstrate that hypergravity can be used as an effective modality for loading skeletal muscle and that subjects can perform squat resistance exercise without becoming motion sick or experiencing illusory motion. The objective of the second study is to determine if the Space Cycle can be used to generate loading conditions that approximate those seen while performing 10 RM squats. A total of 30 subjects (15 male, 15 female) participated in this study, and we are currently analyzing the data at this time.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	There are potentially several unique aspects of the Space Cycle that may have a direct benefit to earth based activities. Currently, it is unclear whether hypergravity may represent a unique loading mode that might enhance training and/or rehabilitation. The unique aspect of hypergravity is that each element is weighted in accordance to its specific gravitational field. In essence, this is like hanging a set of bar bells on every atom/molecule in the body. This distribution of load may be quite different than that of placing weights on a given body structure whereby there is a concentrated loading on that particular structure. Additionally, hypergravity may represent a unique modality for spinal cord injured patients. It is well known that many of these individuals have a reduced orthostatic tolerance. Some have suggested that exposing such individuals to hypergravity may help in this regard. We hope to explore this avenue of investigation in the coming years. Finally, there may be vestibular aspects of Space Cycle activity that might be of benefit to patients who have balance problems.
Task Progress:	The primary tasks of the past year were to: 1. Continue to refine the Space Cycle so that squat resistance training could be performed. 2. To perform a study in both male and female subjects looking at foot forces at different hypergravity conditions. 3. To determine if hypergravity can be used to produce foot forces equivalent to those produced while performing 10 RM squats at 1 Gz.
Bibliography Type:	Description: (Last Updated: 12/17/2008)
Articles in Peer-reviewed Journals	Yang Y, Kaplan A, Pierre M, Adams G, Cavanagh P, Takahashi C, Kreitenberg A, Hicks J, Keyak J, Caiozzo V. "Space cycle: a human-powered centrifuge that can be used for hypergravity resistance training." Aviat Space Environ Med. , in press March 2006. , Mar-2006